Akoizo Original Practice - Form 2 - Answer key

Division B | Roller Coaster | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Why does a rolling ball reach a lower speed than a sliding block after the same drop?

A. Gravity is weaker on spheres

B. It has less potential energy

C. Some energy goes into spinning

D. It is heavier

Correct: C

2. (2 points) How many scorable runs may a team complete?

Unit: Testing and competition day.
Worked-case data: Target 52 s. Practice run: 50.6 s. The calibration table says each screw turn adds 0.4 s. What should the team do?
Separate investigation observations:
Test week 1: Loop failed 3 of 10 runs.
Cause: Ball hit a raised joint just before the loop.
After sanding joint: Loop worked 10 of 10.

A. Up to two

B. One

C. Three

D. Unlimited

Correct: A

3. (2 points) The ball stops on a flat section for 4 s, then rolls on. What happens?

A. Timing ended when it stopped; no Time Score

B. Timing continues

C. A 5-point penalty

D. The run is repeated

Correct: A

4. (2 points) Which loop design earns no Loop Score?

Unit: Loops: physics and scoring.
Worked-case data: A loop has a 9.0 cm path radius. What is the ideal minimum release height above the loop bottom, and a practical target?
Separate investigation observations:
Release 24 cm: Ball fell from top of loop in 6 of 10 trials.
Release 28 cm: Ball completed loop 10 of 10.
Loop path radius: 9 cm.

A. A fully enclosed tube loop

B. An open loop with side rails

C. A loop with a labeled POI

D. A loop 12 cm tall

Correct: A

5. (4 points) Why is a tall loop usually worth extra device height?

Unit: Rules and the four-part score.
Worked-case data: Target Time 45 s. A 52 x 26 x 72 cm device runs 44.6 s with both elevators and an 18.4 cm loop. Find the Run Score.
Separate investigation observations:
Design A: 15 cm loop, no elevators, size 30, time 215.
Design B: 20 cm loop, two elevators, size 15, time 210.
Target Time: 45 s for both.

answer: Each cm of loop height earns 3 points, while each cm of extra height costs only 1 Size point. (4 points).

6. (4 points) Why is width often the cheapest dimension to reduce?

Unit: Size and construction.
Worked-case data: Option 1: 55 x 25 x 75 cm with a 22 cm loop. Option 2: 45 x 20 x 65 cm with a 15 cm loop. Compare Size + Loop points.
Separate investigation observations:
Home runs: Consistent 44.7 s.
After car ride: Runs 42.1 s; base slightly twisted.
After leveling: Runs 44.6 s.

answer: Tracks can be folded into a narrow tower, keeping length and height for timing and the loop. (4 points).

7. (4 points) What is the IMA of a wheel and axle with the effort on a 1 cm axle and the load at a 5 cm wheel?

answer: 1 ÷ 5 = 0.2. (4 points).

8. (4 points) Why plan a height budget?

Unit: Energy, friction, and the rolling ball.
Worked-case data: A solid ball drops 0.40 m from rest without losses. Find its speed.
Separate investigation observations:
Predicted speed: 2.4 m/s at the bottom of a 0.40 m drop.
Measured speed: 2.0 m/s.
Track: Three joints along the drop.

answer: Height is the ball’s only energy; the loop, timing sections, and finish all need some of it. (4 points).

9. (4 points) Describe a plan for the 8-minute period.

answer: Set the Target Time adjustment, do one or two practice runs, fine-tune, then complete two scorable runs. (4 points).

10. (4 points) Why is the median of three timers used?

Unit: Hitting the Target Time.
Worked-case data: Calibration: base path 34 s; each slope-screw turn adds 0.8 s; removable spiral adds 9 s. Plan for a Target Time of 46 s.
Separate investigation observations:
Runs at setting A: 41.8, 42.1, 41.9 s.
Runs at setting B: 44.6, 45.4, 44.8 s.
Target Time: 45 s.

answer: It reduces the effect of one timer’s error, giving a more reliable run time. (4 points).

11. (2 points) Define “Point of Intersection (POI)” in the context of Roller Coaster.

answer: Where the loop appears to cross itself. (2 points).

12. (2 points) Which design scores higher?

Supplied synthetic observations:
Design A: 15 cm loop, no elevators, size 30, time 215.
Design B: 20 cm loop, two elevators, size 15, time 210.
Target Time: 45 s for both.

A. Design B

B. Design A

C. They tie

Correct: A

13. (2 points) What explains the faster runs after the car ride?

Supplied synthetic observations:
Home runs: Consistent 44.7 s.
After car ride: Runs 42.1 s; base slightly twisted.
After leveling: Runs 44.6 s.

A. The ball became heavier

B. The Target Time changed

C. The twisted base changed the track slopes

Correct: C

14. (2 points) Define “Elevator” in the context of Roller Coaster.

answer: Mechanism lifting the ball to a higher level. (2 points).

15. (2 points) Define “Rolling kinetic energy” in the context of Roller Coaster.

answer: (7/10)mv^2 for a solid sphere. (2 points).

16. (2 points) Define “Spread” in the context of Roller Coaster.

answer: Variation in times between runs. (2 points).

17. (2 points) Define “Funnel” in the context of Roller Coaster.

answer: Cone where the ball circles before dropping. (2 points).

18. (2 points) Define “Unsupported top” in the context of Roller Coaster.

answer: Ball held only by speed against the track at the top. (2 points).